INCH–POUND

MIL–M–12054B
25 October 1994

SUPERSEDING
MIL–M–12054A
19 November 1952

MILITARY SPECIFICATION

MUSTARD, H (LEVINSTEIN PROCESS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of bis (2–chloroethyl) sulfide, $(ClCH_2CH_2)_2S$, hereinafter referred to as H.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL–H–502 – Hexamine (For Use in Ammunition)
MIL–C–3250 – Cylinder, Gases and Liquids, One–Ton, Type D

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD–ENE–S, Aberdeen Proving Ground, MD 21010–5423 by using the self–addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1365

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STANDARDS

FEDERAL

FED-STD-313 – Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 – Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 – Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

U.S. ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

E6-2-2 – Drum, 55-Gallon, Type B, Assembly and Bill of Material
D6-10-28 – Containers, One Ton; Marking Diagram and Bill of Materials

(Copies are available from Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-D, Aberdeen Proving Ground, MD 21010-5423.)

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PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM 3-250 - Storage, Shipment, and Handling of Chemical Agents and Hazardous Chemicals

(Copies required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

D 83 - Red Lead Pigment (R 1989)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

INTERNATIONAL CIVIL AVIATION ORGANIZATION

"Technical Instructions for the Safe Transport of Dangerous Goods by Air"

(Application for copies should be addressed to Document Sales Unit, International Civil Aviation Organization, 1000 Sherbrooke Street West, Suite 400, Montreal, Quebec, Canada H3A 2R2.)

INTERNATIONAL MARITIME ORGANIZATION

"International Maritime Dangerous Goods Code"

(Application for copies should be addressed to the International Maritime Organization, 4 Albert Embankment, London SE1 75R.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document,

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however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. The material shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Iron (as $FeCl_2$)	----	0.25	4.2.4.1
Acidity (as HCl)	----	0.30	4.2.4.2
Distillation loss	----	3.0	4.2.4.3.1
Purity of specimen	68.0	----	4.2.4.3.2

3.2 Material Safety Data Sheets. Material Safety Data Sheets for potassium hydroxide shall be prepared and submitted by the contractor in accordance with FED−STD−313 (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.4.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Quality conformance inspection.

4.2.1 Lotting.

4.2.1.1 Batch process. A lot shall consist of the H produced from no more than one manufacturing batch. (See 6.5.)

4.2.1.2 Continuous process. A lot shall consist of the H produced by one manufacturer, at one plant, in no more than one work shift, from the same materials, under the same manufacturing conditions, and at the same time.

4.2.2 Sampling.

4.2.2.1 For tests of chemical and physical characteristics. See 6.6 for sampling and testing precautions. Five representative specimens of no less than 1 pound each shall be randomly taken from each lot prior to any filling operation. The specimens shall be placed in separate clean, dry containers labeled to identify the lot from which they were taken.

4.2.2.2 For examination of packaging. Each container shall be examined as specified in 4.2.3.2.

4.2.3 Inspection procedure.

4.2.3.1 For tests of chemical and physical properties. See 6.6 for sampling and testing precautions. Each specimen taken in 4.2.2.1 shall be tested as specified in the applicable portions of 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

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4.2.3.2 For examination of packaging. Each container shall be examined for the characteristics listed below. Failure of any container to conform to all characteristics shall be cause for rejection of the container.

(a) Container correct
(b) Container undamaged
(c) Steel tubing and threaded openings and fittings correct
(d) Luting material correct
(e) Marking correct

4.2.3.3 For test of container leakage. Each filled and closed container shall be tested for leakage as specified in 4.2.4.4. If any H vapors are detected, the container shall be rejected.

4.2.4 Tests. See 6.6 for sampling and testing precautions. Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Blank determinations shall be run in parallel with each test and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Iron. Introduce approximately 10 mL of the specimen into a dry, tared, 600−mL beaker provided with a watch glass cover and weigh the beaker and specimen to the nearest milligram (mg). Cautiously add approximately 20 mL of concentrated nitric acid dropwise from a pipet, agitating the solution frequently until the violent frothing subsides. Add 20 mL more of concentrated nitric acid and heat until the frothing until the frothing ceases and the liquid is pale yellow. Cool slightly, add 15 mL of 70−percent perchloric acid, and heat with frequent agitation until dense fumes of perchloric acid are given off and the solution is light yellow. Cool and then dilute to approximately 150 mL with water. Add an excess of 6 *N* ammonium hydroxide solution and boil for 5 minutes. Filter while warm and wash five times with hot 2−percent ammonium chloride solution. Discard the filtrate. Dissolve the precipitate by pouring 30 mL of hot 3 *N* hydrochloric acid through the filter. Collect the filtrate in a 250−mL beaker or casserole. Using the Zimmermann−Reinhardt method, reduce with stannous chloride and titrate with 0.1 *N* potassium permanganate solution. Calculate the percent iron as ferrous chloride as follows:

$$\text{Percent iron} = \frac{12.7\,AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution used, and
W = Weight of specimen in grams

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4.2.4.2 Acidity. Place approximately 5 mL of the specimen in a dry, tared, 250-mL iodine flask and weigh the flask and specimen to the nearest milligram. Wash down the stopper and sides of the flask with approximately 15 mL of chloroform and then add approximately 50 mL of saturated sodium chloride solution. Stopper the flask, shake well, add ten drops of phenolphthalein indicator, and titrate with 0.1 *N* sodium hydroxide solution. Do not shake while titrating, but roll or rock the flask gently to mix the contents. The end point is defined as the point at which the pink color persists in the upper layer for at least 10 seconds upon gentle rotation of the flask. Run a blank test on the same quantities of chloroform and sodium chloride solution used in the specimen test. Calculate the percent acidity as hydrochloric acid as follows:

$$\text{Percent acidity} = \frac{3.65\,A\,(B - C)}{W} - 0.576\,D$$

where: A = Normality of sodium hydroxide solution,
B = Milliliters of sodium hydroxide solution used to titrate the specimen,
C = Milliliters of sodium hydroxide solution used to titrate the blank,
D = Percent ferrous chloride calculated in 4.2.4.1, and
W = Weight of specimen in grams

4.2.4.3 Purity.

4.2.4.3.1 Vacuum distillation. Place approximately 95 gm of the specimen, weighed to the nearest decigram (dg), in a tared, Claissen flask. Assemble the vacuum distillation apparatus as shown in figure 1, and test for leaks at a pressure of 5 ± 1 mm of Hg. Conduct the distillation at a pressure of 5 ± 1 mm of Hg, allowing a small amount of air to pass through the system. Apply heat gradually to the oil bath and distill the H at a rate of 60 drops per minute until the bath reaches a temperature of 175°C. Hold the bath at this temperature until the rate of distillation falls to 1 drop per minute; then discontinue the distillation. Weigh the distillate and residue. Reserve the distillate in the side-arm test tube for the freezing point determination (4.2.4.3.2). Calculate the percent distillation loss as follows:

$$\text{Percent distillation loss} = \frac{(W - B) - A}{(W - B)}$$

where: A = Weight of distillate in grams,
B = Weight of residue in grams, and
W = Weight of specimen in grams

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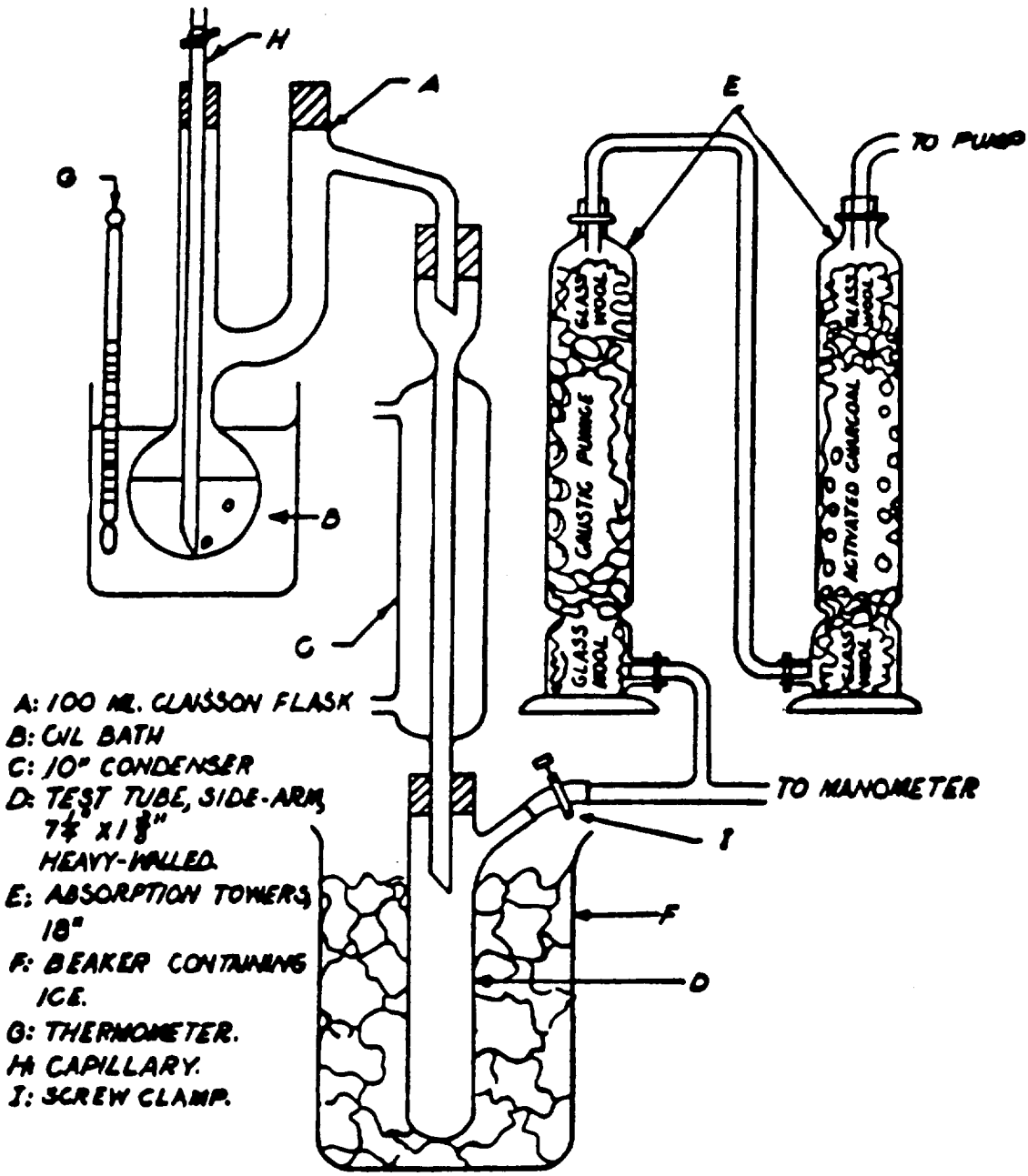


Figure 1. Vacuum distillation apparatus

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4.2.4.3.2 Freezing point. Using the apparatus shown on figure 2 (the thermometer should be first checked against a standardized thermometer for corrections in the appropriate temperature range), fill test tube B with distillate (4.2.4.3.1) to the immersion mark on the thermometer. If an immersion thermometer is not used, fill the tube to 1/2 inch above the thermometer bulb. In no case should less than 25 milliliters (mL) of distillate be used. Assemble the thermometer and stirrer in a cork stopper and place in test tube B as shown on figure 2. The temperature range of the thermometer to be observed should be above the cork. Place test tube B in test tube A and immerse in a bath at approximately 5°C below the anticipated freezing point. Allow the temperature of the distillate to fall to 1.0° ± 0.1°C below the estimated freezing point while stirring. If crystallization does not occur before this point, it must be induced by vigorous stirring or by seeding with previously frozen crystals of the distillate. Record the highest temperature attained during crystallization as the freezing point. If a total immersion thermometer is used, correct for emergent stem. Determine the purity of the distillate by the following formula (it is assumed that for each 1 percent of impurity in the distillate, the freezing point is lowered 0.368°C). The freezing point of pure H is taken as 14.45°C.

$$\text{Percent purity of distillate} = 100 - \frac{(14.45 - A)}{0.368}$$

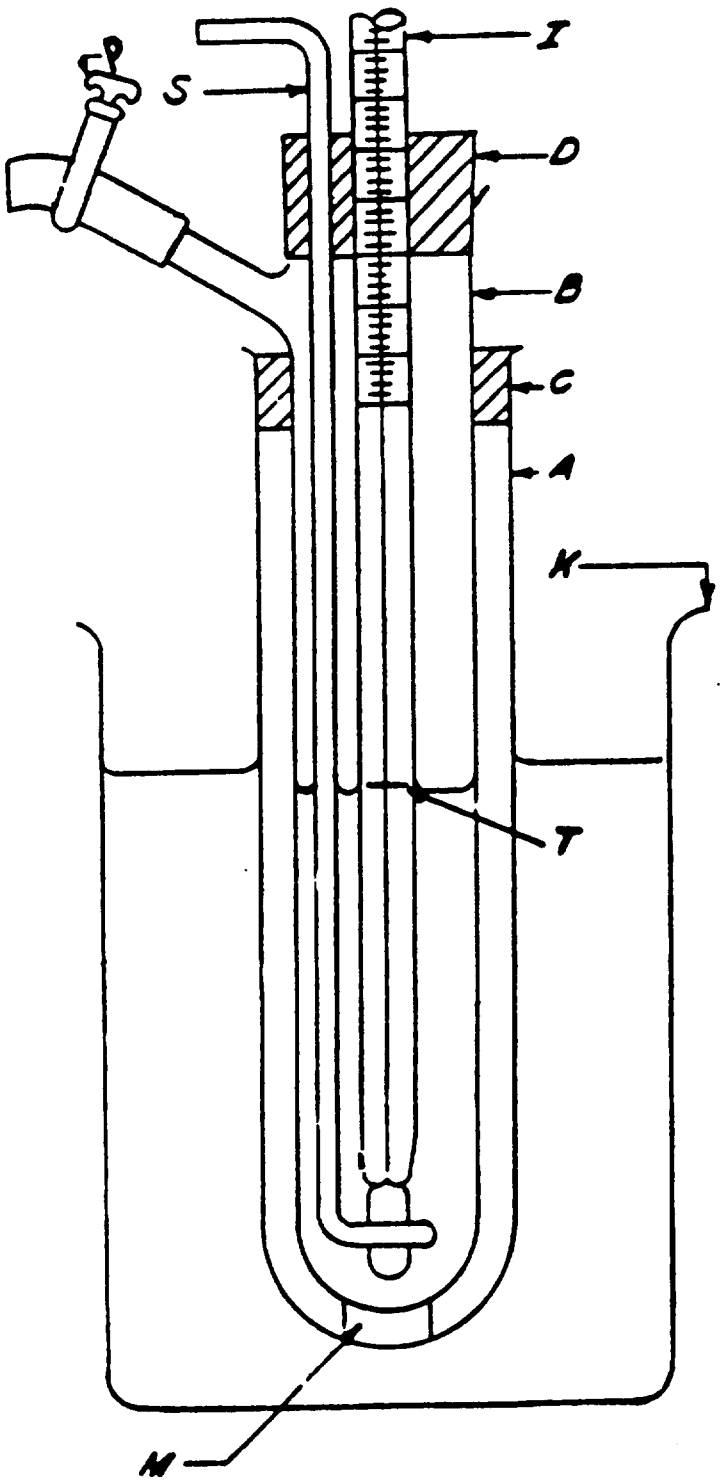
where: A = Freezing point of distillate in °C

Calculate the purity of the specimen as follows:

$$\text{Percent purity of specimen} = \frac{AB}{W}$$

where: A = weight of distillate in grams (4.2.4.3.1),
B = percent purity of distillate, and
W = weight of specimen in grams (4.2.4.3.1)

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A-TEST TUBE 1" DIA. X 6" LONG.

B-TEST TUBE 1 3/8" DIA. X 7" LONG.

C-GOOCH TUBING CONNECTION.

D-STOPPER.

K-BEAKER

I-THERMOMETER GRADUATED TO READ IN TENTHS OF A DEGREE AND ARRANGED TO BE READ ABOVE STOPPER D.

M-CORK SPACER.

P-SCREW CLAMP.

S-STIRRER.

T-THERMOMETER IMMERSION MARK.

Figure 2. Freezing point apparatus

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4.2.4.4 Container leakage. Remove the container to an area which is free of detectable traces of HD as determined by a blue-band detector tube from an M18A1 or M15A1 Detector Kit using the apparatus and method prescribed by the kit. Cover the ends of the container with a shield, leaving a 1-inch diameter opening for sampling the air. Test the air space in the ends for the presence of HD vapors, after a hold period of no less than 30 minutes, using the blue-band detector tube and the method and apparatus from the M18A1 or M15A1 Detector Kit.

5. PACKAGING. Packaging (including marking and labeling) shall be in accordance with the applicable requirements of 49 CFR 171 to 179 and the International Civil Aviation Organization – Technical Instructions for Safe Transportation of Dangerous Goods by Air (ICAO-TDGA) or International Maritime Organization – International Maritime Dangerous Goods Code (IMO-IMDGC), as applicable to the mode of transportation. The packaging shall meet the applicable performance tests specified in the regulations above.

5.1 Packing.

5.1.1 Bulk. A quantity of 170 (+3 or -0) gallons of H shall be packed in a clean, dry, portable tank conforming to Department of Transportation (DOT) 51. Examination, cleaning, drying, and re-examination of the container, steel tubing, and threaded openings and fittings shall be in accordance with the procedures of TM 3-250. Male threaded fittings shall be luted with with red lead paste-in-oil conforming to type II of ASTM D 83. Immediately after filling, all openings shall be securely closed and tested for leakage as specified in 4.2.4.4.

5.1.2 Non-bulk. When H is packed for immediate use, or for transfer to a processing plant, 50 ± 1 gallons may be packed in a 55-gallon steel drum conforming to 1A1 and drawing E6-2-2.

5.1.3 Stabilization. If the H is being packed for storage (not for use at a thickening, filling, purification, or other processing plant) it shall be stabilized by the addition of 1.0 by weight of hexamethylenetetramine conforming to MIL-H-502.

5.2 Marking. In addition to any marking required by the contract or order (see 6.2), each shipping container shall be marked as shown on drawing D6-10-28 (except the symbol shall be H-GAS), in accordance with MIL-STD-129, and in accordance with 49 CFR 171 to 179, ICAO-TDGA, or IMO-IMDGC, as applicable. The "Poison" label shall be applied to each container. If the H has been stabilized as specified in 5.1.3, then the container shall be marked with the symbol "+X" in letters 1 inch high on the valve or bung end of the container.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mustard covered by this specification is intended for use as a chemical agent.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see section 2)
(c) Type of packing required (see 5.1)
(d) Additional marking, if required (see 5.2)

6.3 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent mailing addresses for submissions of data are listed in FED-STD-313.

6.4 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Sampling and testing precautions. This specification requires inspection and use of chemical material which is potentially dangerous to personnel. All applicable safety rules, regulations, and procedures must be followed in the sampling and testing of this material.

6.7 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.9 Subject term (key word) listing.

ammonium chloride
ammonium hydroxide
chloroform
ferrous chloride
hydrochloric acid
lead
mustard (H)
nitric acid
perchloric acid
potassium permanganate
sodium chloride
sodium hydroxide
stannous chloride

Custodian:

Army – EA

Preparing activity:

Army – EA

Project No. 1365-0253

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-12054B	2. DOCUMENT DATE (YYMMDD) 941025

3. DOCUMENT TITLE
MUSTARD, H (LEVINSTEIN PROCESS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
BARRY L. ELLIOTT	(410) 671-3230	DSN 584-3230
c. ADDRESS (Include Zip Code) USA ERDEC ATTN: SCBRD-ENE-S APG, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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